

Gaussian approximation for region-stabilizing functionals and its applications

Joint works with Ilya Molchanov and Riccardo Turin

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Stein's Method: The Golden Anniversary, NUS

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 - Total length of rooted edges
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Motivation - Stabilization

Minimal directed spanning tree (MDST)

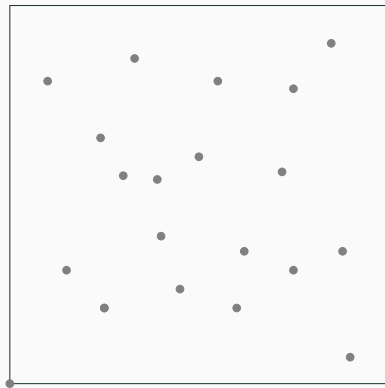


Figure 1: A random MDST in $[0, 1]^2$.

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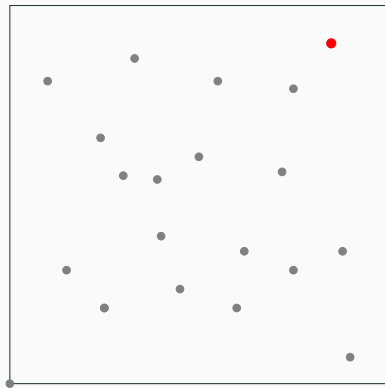


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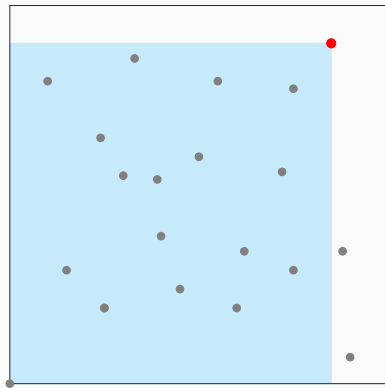


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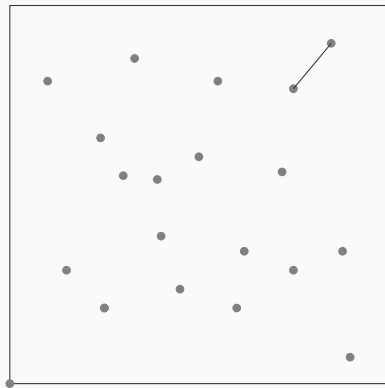


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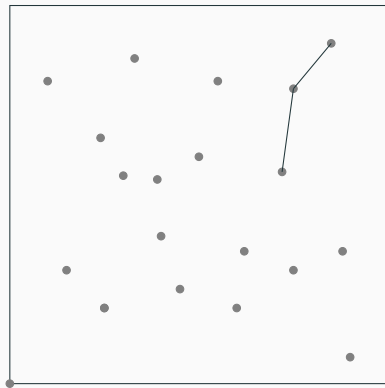


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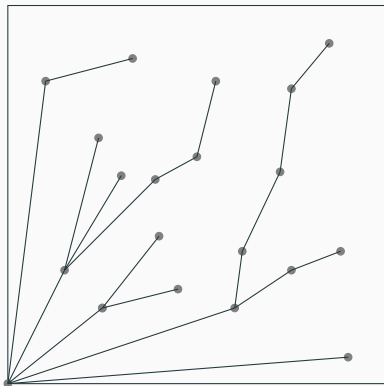


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First considered in Bhatt and Roy (2004) to model transmission or drainage networks, Applications in analysis of **linear programming**, **maxima finding algorithms** ...

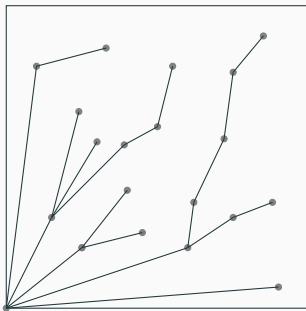


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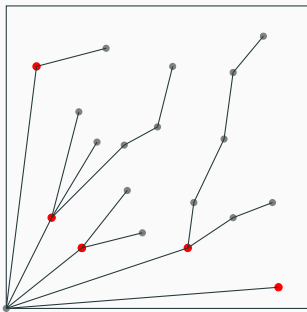


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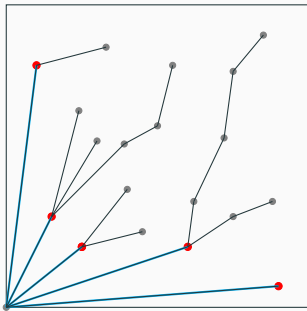


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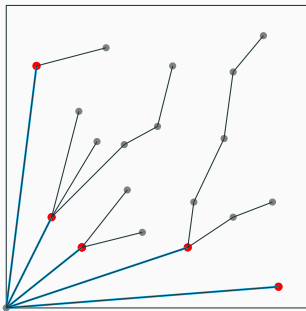


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Almost independent sum; one anticipates **Gaussian fluctuations** of H_s .

Examples

On $[0, 1]^d$, let $H_s = \#$ minimal points in $\mathcal{P}_s$.

$$H_s = \sum_{x \in \mathcal{P}_s} \xi_s(x, \mathcal{P}_s) = \sum_{x \in \mathcal{P}_s} \mathbf{1}(x \text{ is minimal in } \mathcal{P}_s).$$

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As other examples, one can consider **# isolated vertices** or **total edge length** in a **random geometric graph** on $\mathcal{P}_S$ with cutoff radius $\rho > 0$, and many more ...

Distances

The **Wasserstein distance** between real-valued X and Y is given by

$$d_W(X, Y) := \inf \mathbb{E}|X - Y|,$$

where the infimum is taken over all couplings of X and Y .

The **Kolmogorov distance** is defined as

$$d_K(X, Y) := \sup_{t \in \mathbb{R}} |\mathbf{P}\{X \leq t\} - \mathbf{P}\{Y \leq t\}|.$$

Goal: For each s , provide bounds on

$$d_W\left(\frac{H_s - \mathbf{E}H_s}{\sqrt{\mathbf{Var} H_s}}, N\right) \quad \text{and} \quad d_K\left(\frac{H_s - \mathbf{E}H_s}{\sqrt{\mathbf{Var} H_s}}, N\right).$$

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Works for # min. points only for a particular type of domain $\mathbb{X}$.

Application to minimal points

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$$\mathbb{X} = \{x \in (-\infty, 0]^d : F(x) \leq 1\},$$

for some $F \geq 0$ with $F(0) < 1$ and which is **strictly decreasing** coordinatewise, with sufficient smoothness.

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This doesn't apply to $[0, 1]^d$.

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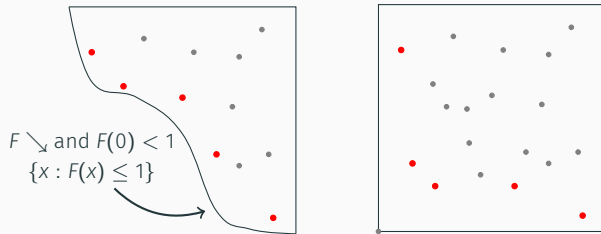


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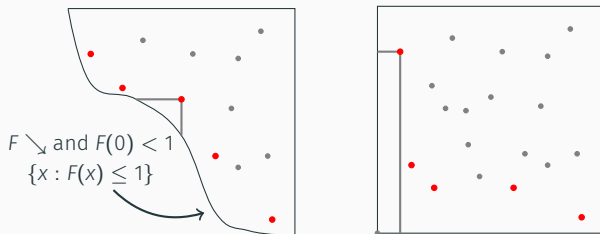


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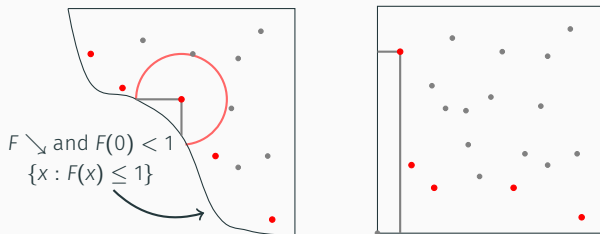


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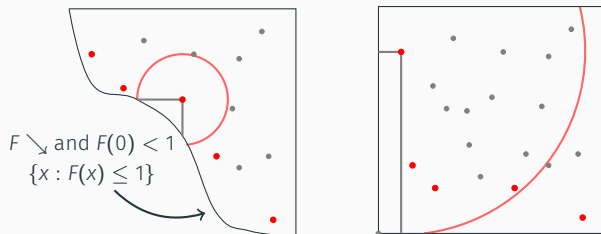


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An extension to general stabilization regions

Assumptions - general regions

Recall, $(\mathbb{X}, \mathcal{F})$ is Borel and $\mathcal{P}_s$ be a Poisson process with intensity $s\mathbb{Q}$ where $\mathbb{Q}$ is σ -finite. Let $\mathbf{N}$ be the family of σ -finite counting measures μ on $\mathbb{X}$.

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(A1) **Stabilization region**: For all $s \geq 1$, there exists a **set-valued** map $R_s : \mathbb{X} \times \mathbf{N} \rightarrow \mathcal{F}$ such that for $x \in \mu$,

$$\xi_s(x, \mu) = \xi_s(x, \mu \cap R_s(x, \mu));$$

+ some natural conditions on ξ_s and R_s + measurability.

Let $r_s : \mathbb{X} \times \mathbb{X} \rightarrow [0, \infty]$ be a non-zero function such that

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(A2) There exists $p \in (0, 1]$ such that, for $\mu \in \mathbf{N}$ with $\mu(\mathbb{X}) \leq 7$,

$$\left\| \xi_s(x, \mathcal{P}_s + \delta_x + \mu) \right\|_{4+p} \leq M_{s,p}(x), \quad s \geq 1, x \in \mathbb{X},$$

Main result

General Bound (B. and Molchanov, 2022)

Assume that $(\xi_s)_{s \geq 1}$ satisfy conditions **(A1)**, **(A2)** and let $H_s = \sum_{x \in \mathcal{P}_s} \xi_s(x, \mathcal{P}_s)$. Then, writing $\mathbb{Q}f := \int f(x) \mathbb{Q}(dx)$,

$$d_W \left(\frac{H_s - \mathbb{E}H_s}{\sqrt{\text{Var } H_s}}, N \right) \leq C \left[\frac{\sqrt{s \mathbb{Q}f_\beta^2}}{\text{Var } H_s} + \frac{s \mathbb{Q}((\kappa_s + g_s)^{2\beta} G_s)}{(\text{Var } H_s)^{3/2}} \right];$$

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and a similar bound holds for the Kolmogorov distance.

Here

$$\kappa_s(x) = \mathbf{P} \{ \xi_s(x, \mathcal{P}_s + \delta_x) \neq 0 \} \text{ and } g_s(y) = s \int_{\mathbb{X}} e^{-\zeta r_s(x,y)} \mathbb{Q}(dx)$$

and C, β, ζ are explicit constants depending on p . The functions f_β and G_s are essentially integrals involving

$$\mathbf{P}\{y \in R_s(x, \mathcal{P}_s + \delta_x)\} \text{ and } \mathbf{P}\left\{ \{y_1, y_2\} \subseteq R_s(x, \mathcal{P}_s + \delta_x) \right\}.$$

Applications

Minimal points

Recall, $\mathcal{P}_s$ is a Poisson process in $[0, 1]^d$ with intensity s and

$$\# \text{ min. points} = \sum_{x \in \mathcal{P}_s} \xi_s(x, \mathcal{P}_s) = \sum_{x \in \mathcal{P}_s} \mathbf{1}(x \text{ is minimal in } \mathcal{P}_s).$$

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Notice,

$$\kappa_s(x) = \mathbf{P}\{\xi_s(x, \mathcal{P}_s + \delta_x) \neq 0\} = e^{-s \text{Vol}[0, x]};$$

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Bai, Devroye, Hwang and Tsai (2005):

$$\text{Var}(\# \text{ min. points}) \sim \log^{d-1} s, \quad s \geq 1.$$

Bound for # Minimal points

CLT for # minimal points (B. and Molchanov, 2022)

Let $d \geq 2$ and F_s be the number of minimal points in an MDST on $\{0\} \cup \mathcal{P}_s$. Then for $s \geq 1$,

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Our bound improves on this rate and provides presumably optimal rates of convergence.

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We Improve this rate for $\alpha = 1$, while for $\alpha \neq 1$, the result is completely new. Rates are presumably optimal.

Digression: the case for $d = 2$

In two dimensions, Penrose and Wade (2004) showed that $\mathcal{L}^\alpha$ converges to the so-called **Dickman distribution** with parameter $2/\alpha$.

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In this case, we prove the following rate of convergence:

(B. and Schulte, 2022+)

$$d_{1,1}(\mathcal{L}^2, D_1) \leq \frac{C}{s}, \quad s \geq 1.$$

Here, $d_{1,1}(X, Y)$ is the so-called *Wasserstein-2* distance, which is slightly weaker than d_W .

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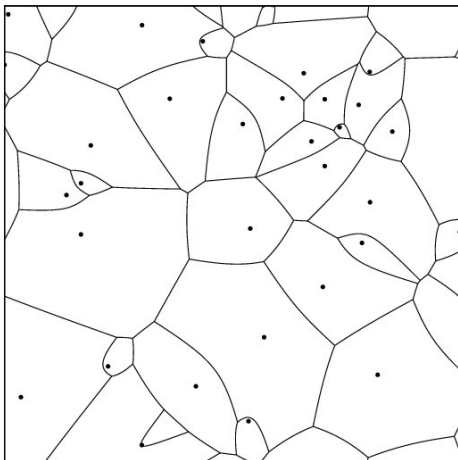


Figure 4: Part of a Johnson-Mehl tessellation, the dots are projections of **exposed seeds** onto $\mathbb{R}^2$ (taken from Bollobás and Riordan (2007))

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Functional of interest:

$$F(\eta) = \sum_{x \in \eta} h_1(x) h_2(t_x) \mathbb{1}\{(x, t_x) \text{ is exposed}\},$$

where $h_1: \mathbb{R}^d \rightarrow [0, \infty)$ and $h_2: \mathbb{R}_+ \rightarrow [0, \infty)$ are measurable weights.

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Example: for $W \subset \mathbb{R}^d$, taking $h_1 = \mathbb{1}_W$ and $h_2(t) = \mathbb{1}(t \leq a)$ gives
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exposed seeds

Assume ν have finite $6d$ -th moment. Let $W_n \subseteq \mathbb{R}^d$ with $\text{Leb}(W_n) \rightarrow \infty$.

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Let $F_n = \#$ exposed seeds in W_n born before time $a \in (0, \infty]$ and θ be Lebesgue. Then $\text{Var}(F_n) \sim \text{Leb}(W_n)$ and

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Previous results needed restrictions on the shape of the domains W_n , we get rid of such assumptions.

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We provide result for general weights, that can be specialised to provide CLT for a variety of related statistics.

Brief overview of the proof of general bound

Tools: Stein's method for normal approximation and second order Poincaré inequality.

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For $x_1, x_2 \in \mathbb{X}$, let

$$D_{y_1} H_s(\mu) = H_s(\mu + \delta_{y_1}) - H_s(\mu), \quad \text{and} \quad D_{y_1, y_2}^2 H_s(\mu) = D_{y_2}(D_{y_1} H_s(\mu))$$

denote the **add-one cost** and the **iterated add-one cost** operators.

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Can be bounded by $\mathbf{P}(y_1 \in R_s(x, \mathcal{P}_s + \delta_x))$ and $\mathbf{P}(\{y_1, y_2\} \subseteq R_s(x, \mathcal{P}_s + \delta_x))$. This and second order Poincaré inequality proves the bound.

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A new avenue to study models with long-range interactions using stabilization.

Thank you!

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Our Stabilization regions satisfy

$$\mathbf{P}(D_{y_1}\xi_s(x, \mathcal{P}_s + \delta_x) \neq 0) \leq \mathbf{P}(y_1 \in R_s(x, \mathcal{P}_s + \delta_x));$$

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Combining this with the second order Poincaré inequality gives the result.